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14. ABSTRACT
This Test Operations Procedure provides the Electromagnetic Environmental Effects testing procedures to supplement Military Standard (MIL-STD)-461 and MIL-STD-464 requirements when applied to automotive platforms while on a dynamometer.

15. SUBJECT TERMS
E3; Electromagnetic Environmental Effects; dynamometer; radiated emissions; radiated susceptibility

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure 02-2-730 19 January 2021

DTIC AD No. AD1120603

Electromagnetic Environmental Effects (E3) Dynamometer Operations

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1. SCOPE.

This Test Operations Procedure (TOP) describes standardized procedures to operate an Electromagnetic Environmental Effects (E3) dynamometer with a turntable to meet the Military Standard (MIL-STD)-461F**[1], MIL-STD-461G[2], and MIL-STD-464[3] requirements of placing the Equipment Under Test (EUT) in the worst case mode of operation during E3 testing.

1.1 Purpose.

a. An E3 chassis dynamometer fulfills the need to place automotive wheeled platforms in the worst case mode of operation to determine the maximum radiated emissions and highlight any potential vulnerabilities as required in MIL-STD-461F, MIL-STD-461G, and MIL-STD-464. Previous generations of automotive platforms were not as reliant on sensors and computers, and they were not required to be moving in order to enter a worst case configuration. Current vehicle platforms are increasingly more reliant on their integrated computers and sensors. This requires the vehicle to be moving with all components active in order to fully assess the vehicle's reliability and vulnerabilities in a Radio Frequency (RF) environment.

b. This TOP describes procedures to allow wheeled vehicles to enter the worst case configuration for radiated emissions and radiated susceptibility. These testing methods will ensure testing repeatability, safety, and allow the automotive platform to be properly assessed.

1.2 Limitations.

a. The facility and instrumentation parameters provided in this TOP are exclusive to the U.S. Army Aberdeen Test Center. Other organizations should tailor this TOP based on their own facility and instrumentation capabilities.

b. TOP describes E3 dynamometer operations for wheeled vehicles up to four (4) axles, less than 34,473.02 kilograms (kg) (76,000 pounds (lbs)) and up to 80.5 kilometers per hour (kph) (50 miles per hour (mph)). These procedures cannot be applied to tracked vehicles and are beyond the scope of this TOP.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

Item	Requirement
Anechoic Chamber	Meet the chamber and anechoic requirements in MIL-STD-461G and large enough to support a dynamometer adequate for the EUT.

** Superscript numbers correspond to Appendix C, References.

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Item	Requirement
Dynamometer	Electromagnetic Interference (EMI) hardened to sustain operations without malfunction in the intended chamber RF environment, and meeting the chamber ambient requirements of at least 6 decibel (dB) below the U.S Army ground limit line in MIL-STD-461. 187.5 peak horsepower (hp) per axle capable of supporting up to 4-axle wheeled vehicles under 34,473.02 kg (76,000 lbs) maximum Gross Vehicle Weight (GVW).
Turntable	EMI hardened to sustain operations without malfunction in the intended chamber RF environment, and meeting the chamber ambient requirements of at least 6 dB below the applicable limit line in MIL-STD-461. Capable of rotating (360 degrees) during testing.

2.2 Instrumentation.

Devices for Measuring	Permissible Measurement Uncertainty
Frequency	2 percent
Amplitude for emissions	2 dB
Field strength	5 percent
Turntable position	0.5 degrees

3. REQUIRED TEST CONDITIONS.

3.1 Preparation For Test.

3.1.1 Vehicle Initial Inspection.

a. Visually inspect test vehicle for any possible damage that might affect the operation of the vehicle on the dynamometer.

(1) Verify that the tire treads are substantial (significant tread depth) enough for dynamometer operation. It is highly recommended to have brand new tires on the vehicle.

(2) Properly inflate all tires to the vehicle manufacturers' specification or in accordance with Technical Manuals as applicable.

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(3) Look for any leaks, class II and above, coming from the vehicle.

(4) Check/fill engine oil, and coolant system.

(5) Check all vehicle tires' treads for embedded debris.

(6) Inspect vehicle undercarriage for loose wires, cables, and debris.

(7) Determine if the undercarriage is required to be sprayed down to remove mud.

(8) Inspect the vehicle exhaust system to determine if there are any leaks.

(9) Perform full preventative maintenance checks and services (PMCS) in accordance with the technical manual to validate the test vehicle is in the normal operating condition.

b. Determine the wheelbase and tire diameter dimensions for the vehicle under test utilizing the vehicle physical characteristics data.

(1) Compare the vehicles wheelbase dimensions to the acceptable range of the dynamometer in Table 1.

TABLE 1. DYNAMOMETER WHEELBASE CONFIGURATIONS

FRONT TANDEM centimeters (inches)	AXLE 2-3 centimeters (inches)	REAR TANDEM centimeters (inches)
121.9-152.4 (48-60)	142.2-342.9 (56-135)	121.9-152.4 (48-60)

(2) Compare the tire diameter to be within the range of 83.8-104.1 centimeters (33-41 inches).

3.1.2 Dynamometer Inspection.

a. Verify all dynamometer emergency-stops are operational.

b. Visually inspect all the roller drive belts to verify they are in good working condition and are ready for test.

c. Check all the airlines to verify they are working correctly and not leaking.

d. Verify that a dynamometer torque and wheelbase calibration is performed every six (6) months, or when an issue is found.

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e. Verify that all debris and foreign objects are cleared away from the rollers and removed from the dynamometer platform.

3.1.3 Turntable Inspection.

a. Verify that all visible casters' hardware are tight and in good working condition. All the casters should be greased at least every six (6) months or when issues are noted.

b. Visually inspect the turntable drive wheel(s) to verify that they are in good working condition.

c. Verify the turntable controller accurately identifies the correct position of the turntable by moving the turntable to known positions and verify it is being displayed on the controller.

3.2 Test Controls.

Prior to testing ensure that the following items are verified:

a. The Heating, Ventilation and Air Conditioning (HVAC) systems are operating correctly to maintain the temperature and humidity requirements of paragraphs 3.3.1.a and b.

b. Chamber vehicle exhaust is functioning by inspecting the exhaust fan drive belt, ensuring the embedded chamber honeycomb filter is not clogged and finally by turning the chamber vehicle exhaust fan on.

c. Chamber vehicle cooling fan is operating correctly and blowing air into the vehicle's radiator.

d. The EMI hardened cameras are operational.

e. Inspect the protective-barriers used around the vehicle's wheels to determine if they are in good working order.

f. All equipment that is not supporting the test is removed from the chamber.

g. Perform a baseline chamber ambient measurement prior to a vehicle entering the chamber using the procedure in paragraph 4.6.1 (only required if you are performing radiated emissions).

3.3 Restrictions.

3.3.1 Environmental.

E3 dynamometer tests are conducted in an anechoic chamber in order to reduce the outside RF environment and eliminate the impacts to the test being conducted. Operating a vehicle on a

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dynamometer in an enclosed chamber requires the HVAC system to maintain a proper environmental condition. The desired conditions are as follows:

a. Ambient temperature: 20 to 30 °Celsius (68 to 86 °Fahrenheit).

b. Humidity: 30 to 70 percent.

c. Chamber air is to be exchanged at least 10 times per hour.

3.3.2 Weights.

a. GVW of the test vehicle needs to be less than 34,473 kg (76,000 lbs) to go on the dynamometer.

b. GVW needs to be less than 4536 kg (10,000 lbs) to go on either side of the dynamometer and over top of the basement.

3.3.3 Size.

a. Vehicle needs to fit in the door opening of 6.1 x 4.9 meters (20 x 16 feet) (width x height).

b. The vehicle with deployed antennas, once in the chamber, cannot exceed 6.1 meters (20 feet) in height.

3.3.4 Dynamometer Related Restrictions.

a. Sustained operations of the dynamometer at high speeds will heat up the vehicle tires causing the tread to potentially deform. This is caused by the bending of the tire on the two roller, cradle style dynamometer.

b. Use only wheeled vehicles (not tracked).

c. Speed on the dynamometer is restricted to be less than 80.5 kph (50 mph).

d. Vehicle axles are limited to be no more than four (4) without modification to the vehicle.

4. TEST PROCEDURES.

4.1 Test Preparation.

The following should be performed before the start of the test:

a. Record the vehicle identification information.

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b. Record the tire size, manufacturer, name, and tire identification number.

c. Obtain Vehicle Characteristics data.

Note: If you will be performing radiated emissions, it is strongly encouraged to perform a chamber RF ambient check before you move the vehicle into the chamber (see paragraph 4.6.1). If you will be performing radiated susceptibility using the recorded method, you should perform the RF field strength calibrations before the vehicle enters the chamber (see paragraph 4.8.2).

4.2 Wheel Base Adjustment.

a. Determine the wheelbase dimensions for the vehicle under test using Vehicle Characteristics data. Compare the wheelbase dimension values to the acceptable wheelbase configuration range of the dynamometer.

b. If the vehicle's wheelbase is within the acceptable range, adjust the dynamometer wheelbase configuration to match the vehicle.

4.3 Moving The Vehicle Onto The Dynamometer.

a. Verify the dynamometer power is off and the dynamometer rollers are locked.

b. Move the turntable to 0 degree position.

c. Raise the dynamometer pneumatic wheel lifting bars to allow the vehicle to be moved into place.

d. Move the vehicle into place above the dynamometer roller cradle with each vehicle wheel directly in the center of the dynamometer roller cradle.

e. Lower the wheel lifting bar.

f. Connect the chamber exhaust hose to the vehicle exhaust system.

g. Verify vehicle cooling fan is placed at proper height to cool the engine (if applicable).

4.4 Vehicle Strap Down Procedure

a. Nylon straps rated for the maximum forces possible from the vehicle, as opposed to metal chains, should be used to reduce any RF interactions during radiated emissions and radiated susceptibility testing.

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b. A minimum of two straps in the front and two straps in the rear of the vehicle should be used and placed as close to the horizontal plane of the dynamometer as possible. This helps to reduce unnecessary stress on the wheel to roller interface leading to higher parasitic losses.

c. Place a non-conductive protective barrier between personnel and the moving rollers/wheels to reduce the risk of being injured in this dangerous area when the dynamometer is operating.

4.5 Operating A Vehicle On The Dynamometer.

a. Verify that the dynamometer wheel lifting bar is lowered.

b. Verify the vehicle is strapped down properly as specified in paragraph 4.4.

c. Verify that the non-conductive protective barriers between personnel and the moving rollers/wheels are present.

d. Turn the dynamometer on and unlock the dynamometer rollers.

e. Turn the chamber vehicle exhaust on.

f. Turn the vehicle on and let it idle.

g. With a Carbon Monoxide (CO) monitor, verify that there is no leakage at the vehicle's exhaust pipes, mufflers, or chamber exhaust tubing while idling.

h. Set up the worst case speed and load that will be used for the testing and verify with the CO monitor that there is no CO leakage in the chamber from the vehicle exhaust.

(1) If the dynamometer hasn't been calibrated, perform the dynamometer calibration in section 4.6 and monitor the CO during the calibration before this CO measurement.

(2) If the dynamometer calibration in section 4.6 has been performed, continue the CO worst case leakage check.

4.6 Dynamometer Calibrations.

a. With the vehicle on the dynamometer, perform the dynamometer warmup routine for 15 minutes to get all the parasitic losses to a constant level.

b. Perform the parasitic losses measurements with the vehicle on the dynamometer whenever the following conditions are encountered:

(1) Whenever the vehicle is changed or if the payload of the test vehicle is adjusted while on the dynamometer.

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(2) Whenever the vehicle restraints are modified.

(3) Whenever the vehicle tire pressure is modified.

c. Perform a dynamometer coast down check to verify that the load applied by the dynamometer is correct.

4.7 Radiated Emissions Testing.

The following radiated emissions testing should be performed to quantify the RF signature of the vehicle under test in its worst case configuration. These radiated emissions could cause issues with communications equipment that would not have been seen in the stationary configuration.

4.7.1 Chamber Ambient Measurement.

a. Turn the vehicle and all subsystems off and shift the vehicle into neutral.

b. Turn the power on to the dynamometer.

c. Run a chamber ambient check using the procedures in MIL-STD-461 subtest RE102 to verify a 6 dB threshold below the U.S Army ground limit line can be achieved.

(1) Vehicle stationary with dynamometer on.

(2) Constant vehicle speed of 24.1 kph (15 mph).

d. Correct any ambient noise issues that are encountered.

(1) Check all items in the vehicle to ensure they are off.

(2) Check to ensure all potential radiators in the chamber are removed.

(3) As a last resort, remove the vehicle from the chamber and check to see if there is something that is always on in the vehicle (if you didn't perform this before the vehicle entered the chamber).

4.7.2 Radiated Emissions Testing Modes.

a. Vehicle stationary with all systems on without dynamometer loading.

b. Vehicle moving with all systems on at 16.1 kph (10 mph) with the following vehicle simulation loading control:

(1) 0 degree grade.

(2) 0 kph/mph headwind.

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c. Vehicle moving with all systems on at 32.2 kph (20 mph) with the following vehicle simulation loading control:

(1) 0 degree grade.

(2) 0 kph/mph headwind.

Note: The speeds in paragraphs 4.7.2b and c can be modified. The rationale for these numbers should be a slower and faster speed to determine if there is a correlation with the radiated emissions to the speed. If a correlation is found, additional speeds may be needed.

4.7.3 Testing Locations To Include (But Not Limited To).

a. Engine compartment (roadside, curbside, and front).

b. Wheel wells.

c. External wiring harnesses.

4.8 Radiated Susceptibility Testing.

The following radiated susceptibility testing should be performed in order to determine the vulnerability of the vehicle under test to external radiated RF signals in their worst case configuration of moving while under load. By having the vehicle move under load, equipment and sensors that would otherwise be inactive in the stationary configuration, would be exercised to determine their vulnerability.

4.8.1 Test Setup (In Addition To The MIL-STD-464/461 Requirements).

a. Front of vehicle set up to be 0 degrees on the turntable.

b. Radiating antenna positioned at 0 degrees

c. Monitoring RF probe set up in front of the test location.

d. Vehicle data bus remoted into the control room using data bus to fiber solutions.

4.8.2 Radiated Susceptibility Recorded Method (The Preferred Method).

Calibration with the vehicle removed from the chamber.

a. Set up the recorded method using Figure 1 as a guide.

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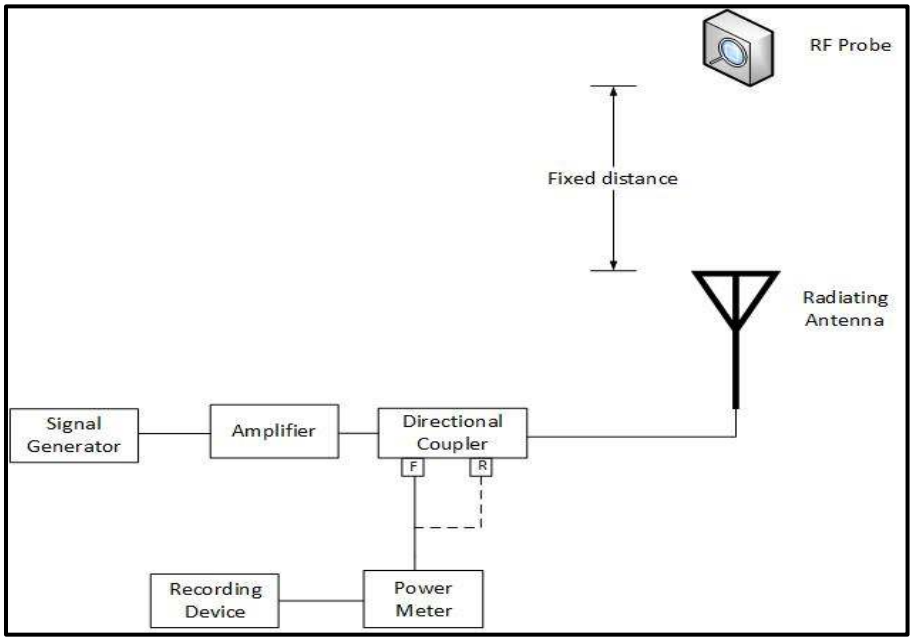


Figure 1. Recorded method calibration setup.

b. Set up a RF probe in the test location.

c. Determine the fixed distance from the antenna to the field probe.

d. From the radiating antenna setup at 0 degrees, radiate a test frequency to the specified test limit.

e. With the inline directional coupler to the radiating antenna, measure and record the forward power, the antenna distance from the field probe, antenna polarity, and the antenna height at the test frequency.

f. Repeat the steps in paragraphs 4.8.2.a through 4.8.3.e for each test frequency including both horizontal and vertical polarity.

4.8.3 Testing With The Vehicle Setup In The Chamber.

a. Setup as depicted in Figure 2.

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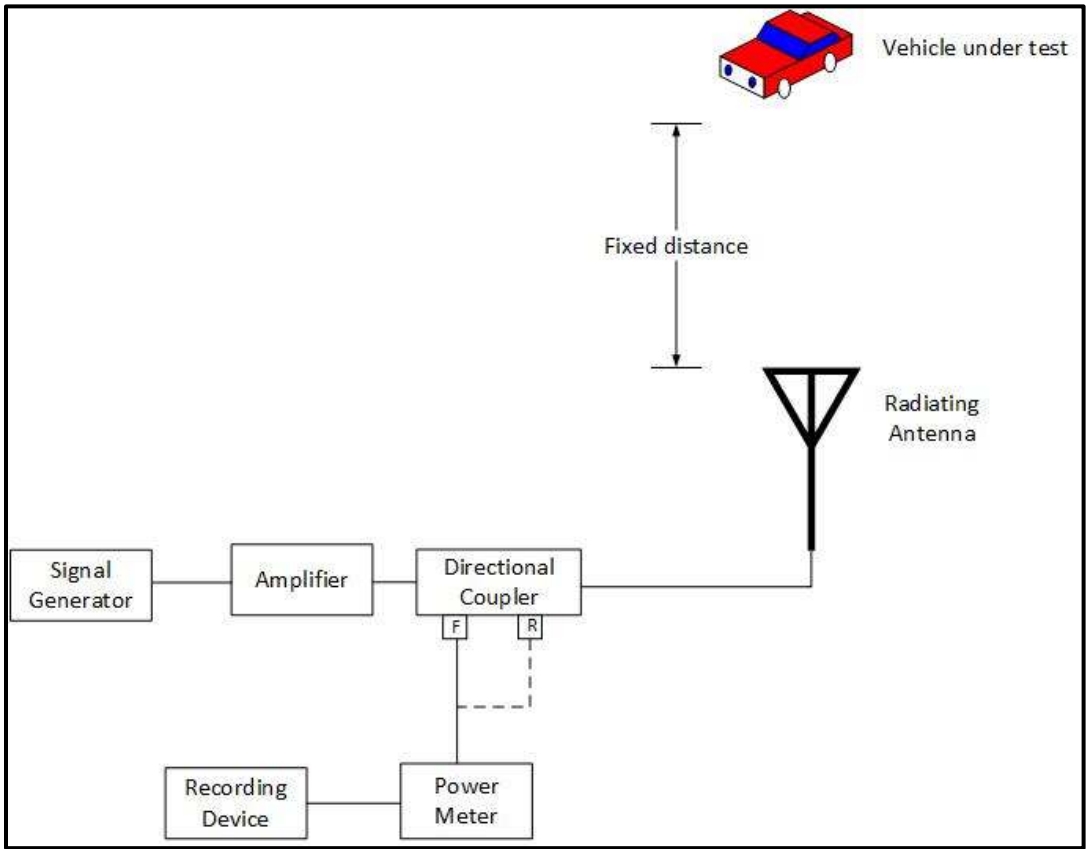


Figure 2. Recorded method testing setup.

b. Using the recorded information for each frequency determined in paragraph 4.8.2, radiate each frequency and polarity at the EUT.

4.8.4 Testing Modes.

The modes specified below are recommended modes and only one may need to be selected based on time and funding. If this is the case, the dynamic speed test is the preferred testing mode due to the most information gained.

a. Stationary. The stationary testing mode is useful to focus on and assess all the onboard systems to determine the vulnerabilities to the radiated signals.

(1) Vehicle on and idling.

(2) All supporting equipment on.

(3) Dynamometer setup for stationary operations.

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b. Constant Speed. The constant speed test is useful to determine vulnerabilities to automotive sensors such as wheel speed. By having a constant speed, you can more easily determine if there is any vulnerability.

(1) Dynamometer setup for a constant power test with 10 hp per axle.

(2) Vehicle will accelerate to 16.1 kph (10 mph) and reach a constant state. This speed can be adjusted but has been selected to not use a lot of fuel but fast enough for the assessment.

(3) Radiated susceptibility testing per MIL-STD-461/464 will commence.

c. Dynamic Speed. Dynamic speed testing is useful to check on any vulnerabilities to the automatic transmission. By having it switch between gears, you can determine if there are any interactions with the RF fields.

(1) Dynamometer setup for a constant power test with 20 hp per axle.

(2) While radiating each frequency, the vehicle will accelerate from 8.0 to 32.2 kph (5 to 20 mph), or above a known vehicle powertrain's shifting point, and reach a constant speed for 5 seconds and then decelerate back to 8.0 kph (5 mph).

(3) Repeat the steps in paragraphs 4.8.4.c(1) and (2) for each test frequency.

4.8.5 Testing Locations To Include (But Not Limited To).

a. Engine compartment (roadside, curbside, and front).

b. Wheel wells.

c. External wiring harnesses.

5. DATA REQUIRED.

a. During radiated emissions testing the following data are commonly recorded:

(1) Chamber ambient temperature.

(2) Parasitic losses graph.

(3) Emissions antenna location.

(4) Emissions antenna polarity.

(5) Frequency vs. signal strength information.

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b. During radiated susceptibility testing the following data are commonly recorded with the corresponding Global Positioning System time stamp:

(1) Torque.

(2) Engine revolutions per minute.

(3) Acceleration.

(4) Speed.

(5) The data bus information from the vehicle.

(6) Turntable position information.

(7) Parasitic losses graph.

(8) Test frequency.

(9) Test amplitude.

(10) Radiating antenna polarity.

(11) Radiating antenna location.

6. PRESENTATION OF DATA.

An example chamber ambience plot is shown in Figure 3.

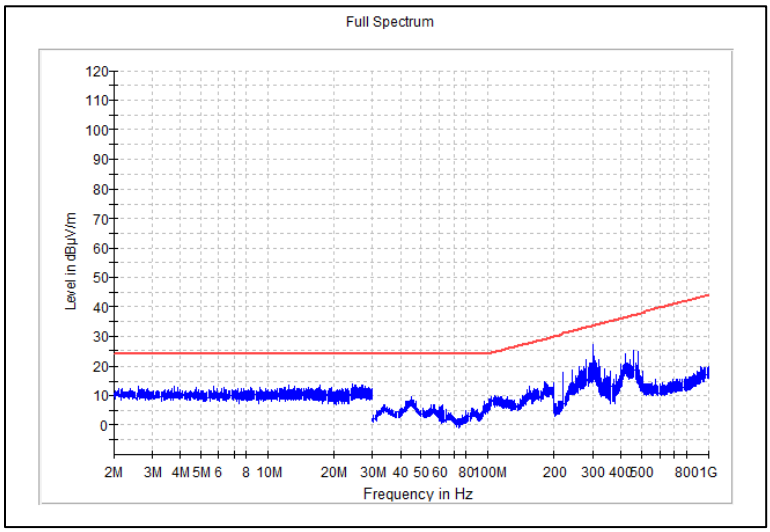


Figure 3. Chamber ambience example.

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APPENDIX A. GLOSSARY.

Term	Definition
Ambient	Relating to the immediate surroundings of something.
Anechoic	Free from echo.
Chassis	The base frame of a motor vehicle or other wheeled conveyance.
Pneumatic	Containing or operated by air or gas under pressure.
Powertrain	The mechanism that transmits the drive from the engine of a vehicle to its axle.
Wheelbase	The distance between the steering (front) axle and the center point of the driving axle group

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APPENDIX B. ABBREVIATIONS.

ATEC	U.S. Army Test and Evaluation Command
CO	carbon monoxide
dB	decibel
E3	Electromagnetic Environmental Effects
EMI	Electromagnetic Interference
EUT	equipment under test
GVW	Gross Vehicle Weight
hp	horsepower
HVAC	heating, ventilation, and air conditioning
kg	kilogram
kph	kilometers per hour
lb	pounds
MIL-STD	Military Standard
mph	miles per hour
RF	radio frequency
TOP	Test Operations Procedure

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APPENDIX C. REFERENCES.

1. MIL-STD-461F, Department of Defense Interface Standard Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, 10 December 2007.

2. MIL-STD-461G, Department of Defense Interface Standard Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, 11 December 2015.

3. MIL-STD-464C, Department of Defense Interface Standard Electromagnetic Environmental Effects Requirements for Systems, December 2010.

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APPENDIX D. APPROVAL AUTHORITY.

CSTE-CI 19 January 2021

MEMORANDUM FOR

Commander, U.S. Army Operational Test Command
Director, U.S. Army Evaluation Center
Commanders, ATEC Test Centers
Technical Directors, ATEC Test Centers

SUBJECT: Test Operations Procedure 02-2-730, Electromagnetic Environmental Effects (E3) Dynamometer Operation, Approved for Publication

1. Test Operations Procedure (TOP) 02-2-730, Electromagnetic Environmental Effects (E3) Dynamometer Operation, has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP describes standardized procedures to operate an E3 dynamometer with a turntable to meet the Military Standard (MIL-STD)-461F, MIL-STD-461G, and MIL-STD-464 requirements of placing the equipment under test in the worst case mode of operation during E3 testing. These testing methods will ensure testing repeatability, safety, and allow the automotive platform to be properly assessed.

3. This document is approved for publication and has been posted to the Reference Library of the ATEC Vision Digital Library System (VDLS). The VDLS website can be accessed at https://vdls.atc.army.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army Test and Evaluation Command (CSTE-CI), 6617 Aberdeen Boulevard-Third Floor, Aberdeen Proving Ground, MD 21005-5001; or e-mailed to usarmy.apg.atec.mbx.atec-standards@mail.mil.

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MICHAEL J. ZWIEBEL
Director, Directorate for Capabilities Integration (DCI)

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